Power input (USB−C) and regulators

File: lcd_dual_power.kicad_sch

HDMI receiver LT8619C

File: lcd_dual_hdmi.kicad_sch

AT070TN92 panel, bias supplies, VCOM, backlight

File: lcd_dual_panel.kicad_sch

STM32F072 controller and GT915 touch

File: lcd_dual_mcu.kicad_sch

AT070TN92 7'' 800x480 TTL panel driver: HDMI (LT8619C) + GT915 touch −> USB HID (STM32F072).

Power: USB−C 5 V (from Raspberry Pi USB port, ~0.7 A max with backlight at 100 %).

U10: STM32F072CBT6 or CH32X035C8T6 (same LQFP−48 footprint). SW2 = CH32 USB bootloader (D+ high at power−up).

Panel power−up order (MCU controlled): DVDD −> RESET −> VGL/AVDD −> VGH −> data −> backlight.

J3 pad numbers equal AT070TN92 FPC pin numbers; use a TOP−contact 0.5 mm 50−pin FPC connector.

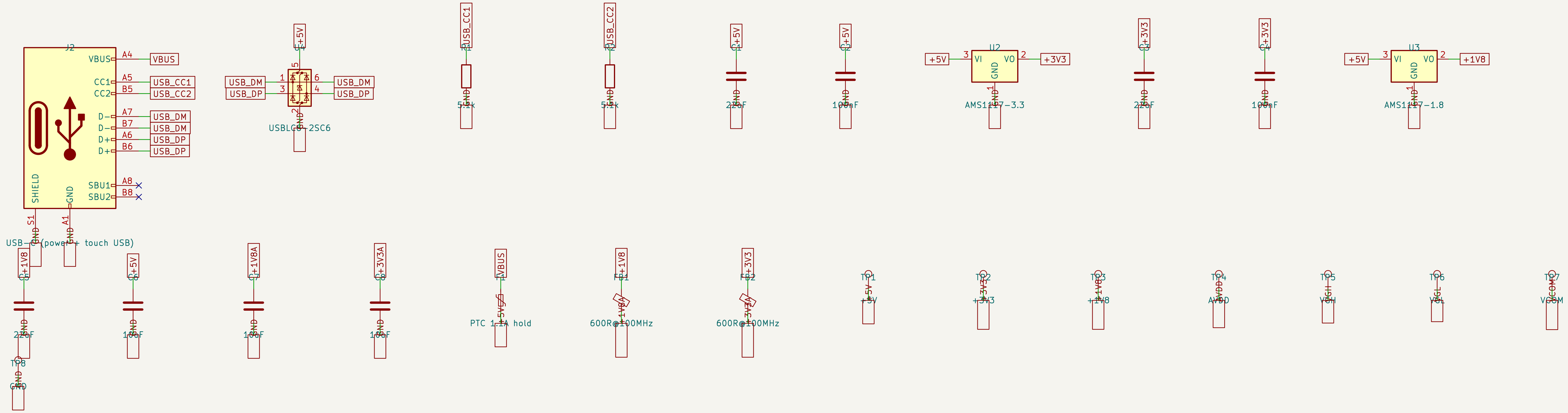
VCOM (pins 6/46) is adjustable with RV1 (2.55..4.5 V): trim for minimum flicker.

LT8619C HDMI−>RGB888, TPS61040 bias, TPS61165 backlight, STM32F072 USB touch

Sheet: /
File: lcd_dual.kicad_sch

Title: AT070TN92 HDMI + USB−touch driver board

Size: A4 Date: 2026−10−02 **Rev: 1.0**
KiCad E.D.A. kicad−cli 7.0.11+dfsg−1build4 Id: 1/5


J2
VBUS
A4
VBUS
CC1
A5
USB_CC1
CC2
B5
USB_CC2
D−
A7
USB_DM
D−
B7
USB_DM
D+
A6
USB_DP
D+
B6
USB_DP
SBU1
A8
SBU2
B8
SHIELD
GND
S1
A1
GND
GND
USB−C (power + touch USB)
+5V
U4
5
USB_DM
1
6
USB_DM
USB_DP
3
4
USB_DP
2
GND
USBLC6−2SC6
USB_CC1
R1
GND
5.1k
USB_CC2
R2
GND
5.1k
+5V
C1
GND
22uF
+5V
C2
GND
100nF
U2
+5V
3
VI
VO
2
+3V3
GND
1
GND
AMS1117−3.3
+3V3
C3
GND
22uF
+3V3
C4
GND
100nF
U3
+5V
3
VI
VO
2
+1V8
GND
1
GND
AMS1117−1.8
+1V8
C5
GND
22uF
TP8
GND
GND
+5V
C6
GND
10uF
+1V8A
C7
GND
10uF
+3V3A
C8
GND
10uF
VBUS
F1
+5V
PTC 1.1A hold
+1V8
FB1
+1V8A
600R@100MHz
+3V3
FB2
+3V3A
600R@100MHz
TP1
+5V
+5V
TP2
+3V3
+3V3
TP3
+1V8
+1V8
TP4
AVDD
AVDD
TP5
VGH
VGH
TP6
VGL
VGL
TP7
VCOM
VCOM

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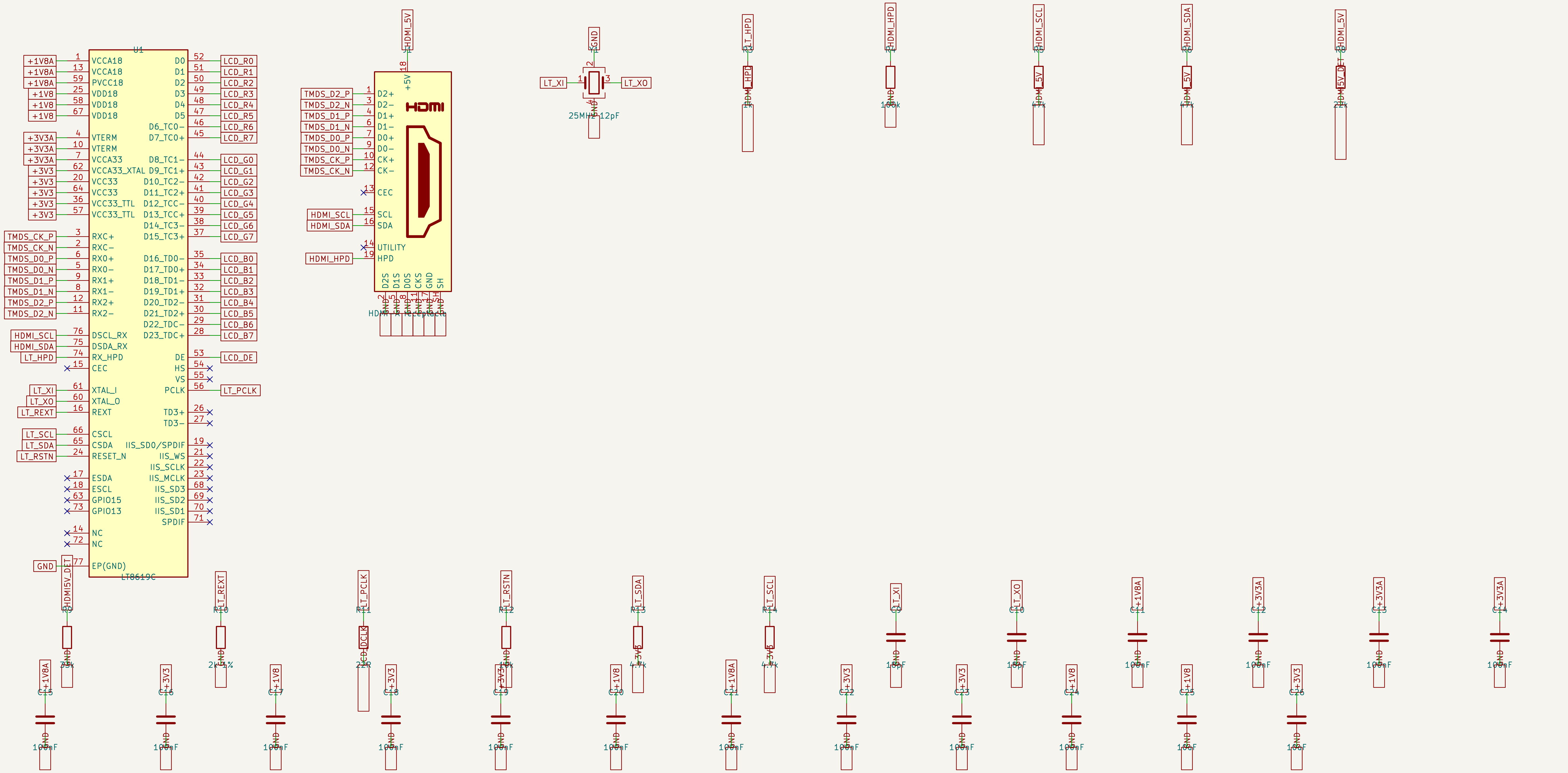
AT070TN92 HDMI + USB touch driver
Sheet: /Power input (USB−C) and regulators/
File: lcd_dual_power.kicad_sch

Title: Power input (USB−C) and regulators

Size: A3 Date: 2026−10−02 **Rev: 1.0**
KiCad E.D.A. kicad−cli 7.0.11+dfsg−1build4 Id: 2/5


U1
LT8619C
VCCA18
PVCC18
VDD18
VTERM
VCCA33
VCCA33_XTAL
VCC33
VCC33_TTL
RXC+
RXC−
RX0+
RX0−
RX1+
RX1−
RX2+
RX2−
DSCL_RX
DSDA_RX
RX_HPD
CEC
XTAL_I
XTAL_O
REXT
CSCL
CSDA
RESET_N
ESDA
ESCL
GPIO15
GPIO13
NC
EP(GND)
D0
D1
D2
D3
D4
D5
D6_TC0−
D7_TC0+
D8_TC1−
D9_TC1+
D10_TC2−
D11_TC2+
D12_TCC−
D13_TCC+
D14_TC3−
D15_TC3+
D16_TD0−
D17_TD0+
D18_TD1−
D19_TD1+
D20_TD2−
D21_TD2+
D22_TDC−
D23_TDC+
DE
HS
VS
PCLK
TD3+
TD3−
IIS_SD0/SPDIF
IIS_WS
IIS_SCLK
IIS_MCLK
IIS_SD3
IIS_SD2
IIS_SD1
SPDIF
+1V8A
+1V8
+3V3A
+3V3
TMDS_CK_P
TMDS_CK_N
TMDS_D0_P
TMDS_D0_N
TMDS_D1_P
TMDS_D1_N
TMDS_D2_P
TMDS_D2_N
HDMI_SCL
HDMI_SDA
LT_HPD
LT_XI
LT_XO
LT_REXT
LT_SCL
LT_SDA
LT_RSTN
GND
LCD_R0
LCD_R1
LCD_R2
LCD_R3
LCD_R4
LCD_R5
LCD_R6
LCD_R7
LCD_G0
LCD_G1
LCD_G2
LCD_G3
LCD_G4
LCD_G5
LCD_G6
LCD_G7
LCD_B0
LCD_B1
LCD_B2
LCD_B3
LCD_B4
LCD_B5
LCD_B6
LCD_B7
LCD_DE
LT_PCLK
HDMI_5V
+5V
HDMI
D2+
D2−
D1+
D1−
D0+
D0−
CK+
CK−
CEC
SCL
SDA
UTILITY
HPD
D2S
D1S
D0S
CKS
GND
SH
HDMI_HPD
Y1
25MHz 12pF
R3
R4
R5
R6
R7
R8
100k
47k
22k
HDMI5V_DET
R9
R10
R11
R12
R13
R14
2k 1%
22R
10k
4.7k
LCD_PCLK
C9
C10
C11
C12
C13
C14
C15
C16
C17
C18
C19
C20
C21
C22
C23
C24
C25
C26
18pF
100nF
10uF

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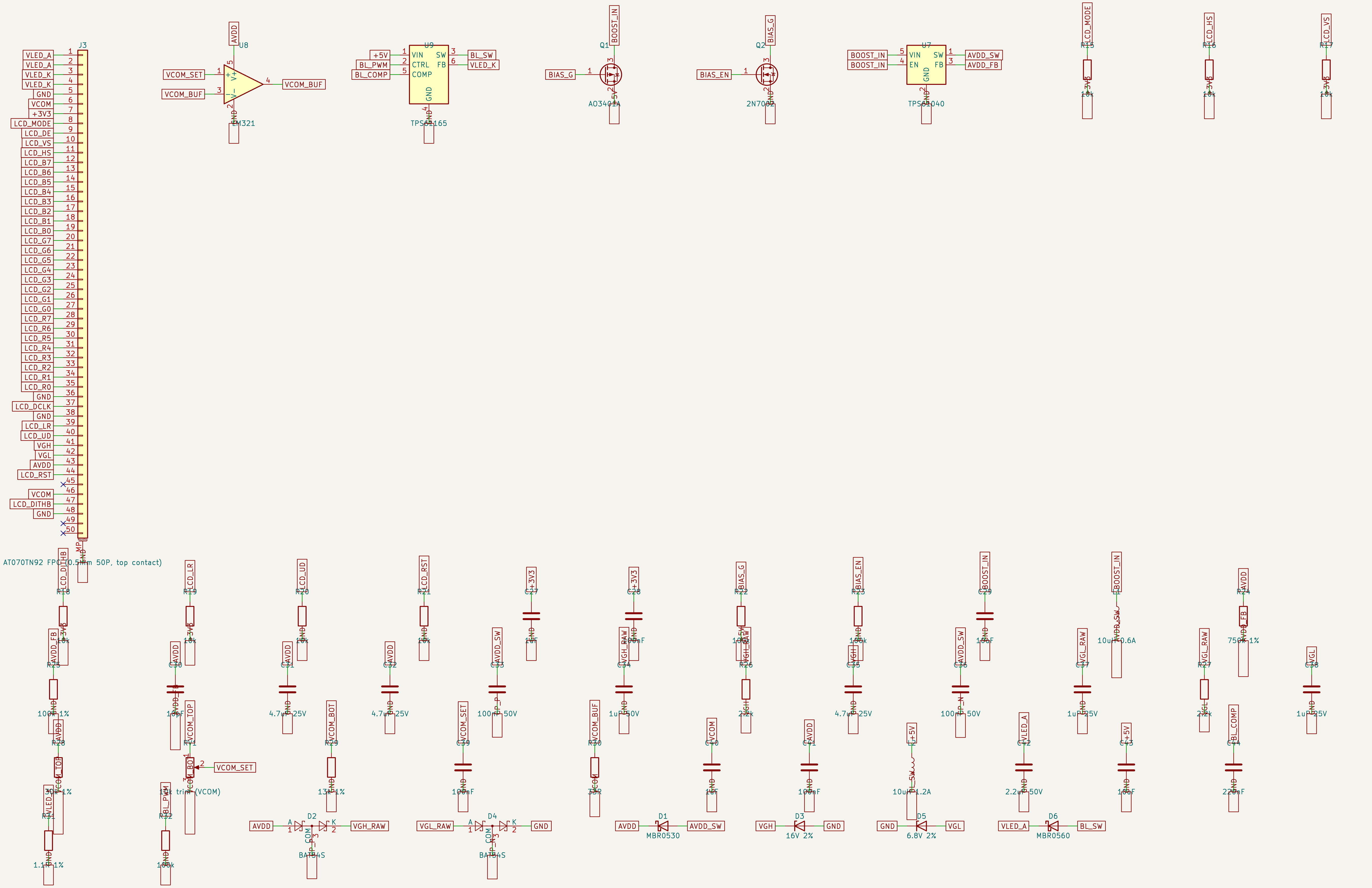
AT070TN92 HDMI + USB touch driver
Sheet: /HDMI receiver LT8619C/
File: lcd_dual_hdmi.kicad_sch

Title: HDMI receiver LT8619C

Size: A3 Date: 2026-10-02 **Rev: 1.0**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 Id: 3/5


J3
VLED_A 1
VLED_A 2
VLED_K 3
VLED_K 4
GND 5
VCOM 6
+3V3 7
LCD_MODE 8
LCD_DE 9
LCD_VS 10
LCD_HS 11
LCD_B7 12
LCD_B6 13
LCD_B5 14
LCD_B4 15
LCD_B3 16
LCD_B2 17
LCD_B1 18
LCD_B0 19
LCD_G7 20
LCD_G6 21
LCD_G5 22
LCD_G4 23
LCD_G3 24
LCD_G2 25
LCD_G1 26
LCD_G0 27
LCD_R7 28
LCD_R6 29
LCD_R5 30
LCD_R4 31
LCD_R3 32
LCD_R2 33
LCD_R1 34
LCD_R0 35
GND 36
LCD_DCLK 37
GND 38
LCD_LR 39
LCD_UD 40
VGH 41
VGL 42
AVDD 43
LCD_RST 44
45
VCOM 46
LCD_DITHB 47
GND 48
49
50
MP
GND
AT070TN92 FPC (0.5mm 50P, top contact)
AVDD
U8
VCOM_SET
VCOM_BUF
VCOM_BUF
LM321
GND
U9
+5V
BL_PWM
BL_COMP
VIN
CTRL
COMP
SW
FB
BL_SW
VLED_K
GND
TPS61165
BOOST_IN
Q1
BIAS_G
AO3401A
+5V
BIAS_G
Q2
BIAS_EN
2N7002
GND
U7
BOOST_IN
BOOST_IN
VIN
EN
SW
FB
GND
AVDD_SW
AVDD_FB
TPS61040
LCD_MODE
R15
+3V3
10k
LCD_HS
R16
+3V3
10k
LCD_VS
R17
+3V3
10k
LCD_DITHB
R18
+3V3
10k
LCD_LR
R19
+3V3
10k
LCD_UD
R20
GND
10k
LCD_RST
R21
GND
10k
+3V3
C27
GND
1uF
+3V3
C28
GND
100nF
BIAS_G
R22
+5V
100k
BIAS_EN
R23
GND
100k
BOOST_IN
C29
GND
10uF
BOOST_IN
L1
AVDD_SW
10uH 0.6A
AVDD
R24
AVDD_FB
750k 1%
AVDD_FB
R25
GND
100k 1%
AVDD
C30
AVDD_FB
10pF
AVDD
C31
GND
4.7uF 25V
AVDD
C32
GND
4.7uF 25V
AVDD_SW
C33
CP_P
100nF 50V
VGH_RAW
C34
GND
1uF 50V
VGH_RAW
R26
VGH
2.2k
VGH
C35
GND
4.7uF 25V
AVDD_SW
C36
CP_N
100nF 50V
VGL_RAW
C37
GND
1uF 25V
VGL_RAW
R27
VGL
2.2k
VGL
C38
GND
1uF 25V
AVDD
R28
VCOM_TOP
30k 1%
VCOM_TOP
RV1
VCOM_BOT
VCOM_SET
10k trim (VCOM)
VCOM_BOT
R29
GND
13k 1%
VCOM_SET
C39
GND
100nF
VCOM_BUF
R30
VCOM
33R
VCOM
C40
GND
1uF
AVDD
C41
GND
100nF
+5V
L2
BL_SW
10uH 1.2A
VLED_A
C42
GND
2.2uF 50V
+5V
C43
GND
10uF
BL_COMP
C44
GND
220nF
VLED_K
R31
GND
1.1R 1%
BL_PWM
R32
GND
100k
AVDD
D2
CP_P
VGH_RAW
BAT54S
VGL_RAW
D4
CP_N
GND
BAT54S
AVDD
D1
AVDD_SW
MBR0530
VGH
D3
GND
16V 2%
GND
D5
VGL
6.8V 2%
VLED_A
D6
BL_SW
MBR0560

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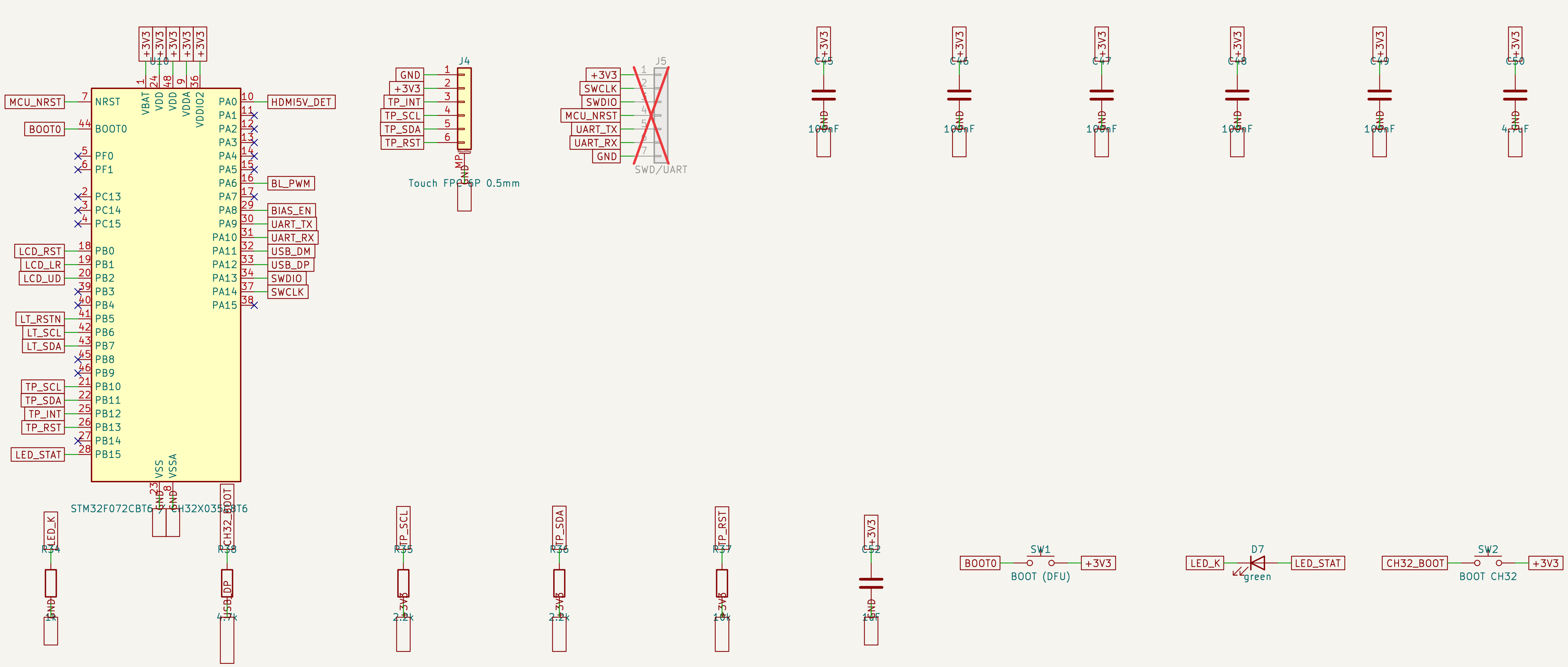
AT070TN92 HDMI + USB touch driver
Sheet: /AT070TN92 panel, bias supplies, VCOM, backlight/
File: lcd_dual_panel.kicad_sch

Title: AT070TN92 panel, bias supplies, VCOM, backlight

Size: A3	Date: 2026-10-02	**Rev: 1.0**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 4/5


U10
STM32F072CBT6 / CH32X035C8T6
+3V3
VBAT
VDD
VDDA
VDDIO2
MCU_NRST
NRST
BOOT0
PF0
PF1
PC13
PC14
PC15
LCD_RST
LCD_LR
LCD_UD
PB0
PB1
PB2
PB3
PB4
LT_RSTN
LT_SCL
LT_SDA
PB5
PB6
PB7
PB8
PB9
TP_SCL
TP_SDA
TP_INT
TP_RST
PB10
PB11
PB12
PB13
PB14
LED_STAT
PB15
VSS
VSSA
GND
PA0
HDMI5V_DET
PA1
PA2
PA3
PA4
PA5
PA6
BL_PWM
PA7
PA8
BIAS_EN
PA9
UART_TX
PA10
UART_RX
PA11
USB_DM
PA12
USB_DP
PA13
SWDIO
PA14
SWCLK
PA15
J4
Touch FPC 6P 0.5mm
MP
J5
SWD/UART
SWCLK
SWDIO
UART_TX
UART_RX
C45
C46
C47
C48
C49
C50
C51
100nF
4.7uF
R33
10k
R34
1k
LED_K
R38
4.7k
CH32_BOOT
R35
2.2k
R36
2.2k
R37
10k
C52
1uF
SW1
BOOT (DFU)
D7
green
SW2
BOOT CH32

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AT070TN92 HDMI + USB touch driver
Sheet: /STM32F072 controller and GT915 touch/
File: lcd_dual_mcu.kicad_sch

Title: *STM32F072 controller and GT915 touch*

Size: A3 | Date: 2026-10-02 | **Rev: 1.0**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | Id: 5/5